



OPEN PLAN

A Design History of the American Office

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CULTURAL HISTORIES
OF DESIGN

BLOOMSBURY

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Chapter 1

Designing Hierarchy

In the 1980 film *9 to 5*, Lily Tomlin, Dolly Parton, and Jane Fonda played three white-collar workers who kidnap their “sexist, egotistical, lying, hypocritical bigot” boss, played by Dabney Coleman. In his absence from the office, they transform the bureaucratic and deeply sexist corporate culture into a more supportive organization with a host of new feminist policies and benefits including flexible hours, job share, equal pay for equal work, and in-office daycare. Over the course of the movie, office design serves as a direct expression of this transformation. At the start of the film, the office is a dull and monochromatic bullpen with rows and rows of tidy desks arranged in a grid, a spatial reflection of the top-down organizational culture of the corporation. By the end of the film the three women had transformed that same office into a new open plan, featuring a looser arrangement of desks in small groupings separated by low brightly colored orange and yellow partitions. Notably, the image of the new open plan in the film also emphasizes the diversity and inclusiveness of the space—the view of the newly designed office prominently features women, workers of color, and even a worker in a wheelchair pulling up to their open plan workstation reinforcing the idea that this workplace is an entirely different one not only spatially, but also culturally. This symbolic intertwining of an inclusive organizational culture and the open plan office design in the film reflects the image of the open plan as a naturally progressive office design solution and an expression of a more egalitarian organizational culture.

This chapter traces these changing ideas about hierarchy and office design from the conventionally designed offices of the mid-twentieth century with its rigid expression of hierarchy and status to the new and supposedly more inclusive open plan. Architects and designers of the period often treated status in office design as a neutral expression of organizational position and status, but I argue that office hierarchies of this era were intertwined with the long history of discrimination in the workplace, particularly in terms of gender, race, and ability. Advocates in the late 1960s viewed the new open plan as a complete

reimagining of organizational culture, management theory, and the office itself, particularly in its rejection of hierarchy and status as organizing structures of the office. Echoing the emerging ideas in management theory of the 1960s, architects and designers promoting the open plan envisioned a future in which workers would have greater ownership over their work and work process, which in turn would benefit the organization by increasing productivity, efficiency, and creativity. Rather than imagining the organization as a system of rationalized boxes reporting up the chain of command (as represented in an organizational chart of the era), new management theory described the modern organization as a flatter system in which all the parts of the organization were important contributors to the larger goal. For proponents of the open plan, the new office concept was the spatial manifestation of this larger transformation in the structure of work that promised a more egalitarian culture.

Even as the open plan projected an image of greater inclusivity, hierarchy was always an integral part of the planning process for the open plan. Though its advocates often described the benefits of worker participation in office planning, the open plan was typically implemented in ways that codified and reinforced the positions, priorities, and preferences of those in power and often left out altogether the experiences and needs of lower level support staff and service workers, who rarely had the opportunity to participate in the design process in a substantive way. Recognizing the long tail of discrimination and exclusion embedded in hiring and promotion practices in American organizations throughout the twentieth century, this system effectively prioritized the voices of predominantly White men in professional roles while ignoring the voices, perspectives, and needs of marginalized workers in lower level positions including women, workers with disabilities, and workers of color. Despite its progressive and countercultural image, the open plan in fact *reinforced* organizational structures and systems of inequality through its design and usage.

Postwar Corporate Hierarchy

The postwar period saw an enormous growth in the design and construction of new corporate offices in the United States.¹ Rather than rehabilitate, remodel, or update their older buildings, American companies of that era invested enormous resources in constructing entirely new office buildings to suit their rapidly expanding organizations. Whether located inside of a towering skyscraper in the central business district of a major city or in a sprawling corporate office in the suburbs, these new offices reflected the organizational priorities and needs of the time. As Louise Mzingo argues, through their design, these new corporate headquarters of the postwar era needed to: serve as an architectural symbol of the organization, impressing investors and banks and conveying their “corporate

trustworthiness”; attract and keep top executives, ensuring that those corporate executives felt a connection to the organization as a whole; and reassure the public that their fears about the excessive power of large corporations were unfounded. According to Mozingo, the corporate architecture of the postwar became the public image of the organization, and a “tangible corporate persona.”² Further, as Alexandra Lange argues, corporate office design of the postwar was not simply “a matter of dressing the old building in a new glass-and-steel wrapper,” it was an “overhaul” of the organizational structure by way of architecture.³

Architects and designers planned the interior space of new corporate offices around the organizational structure in explicit ways. First, they designed these new offices based on the formal departmental divisions, breaking up the organization into its constituent parts as conceptualized by the organizational chart and arranging the various corporate divisions around their functional and hierarchical relationships with other units. Within each department, the division of space reflected the inner workings of that department as well as the functional and hierarchical relationships among workers. Office planners typically provided workers in elevated positions a private office while they placed workers in lower-level positions, particularly those in technical and clerical roles, in an open “bullpen.” Mapping the interior, architects and designers created detailed standards that allocated space, furniture, and accessories based primarily on the organizational status of each worker. The interior standards established for a particular project rationalized the allocation of space and amenities across the organization to reflect differential status from the topmost executives to the lowest-level clerical workers.⁴ In her study of a single organization, sociologist Rosabeth Moss Kanter found a “clear system of stratification” in the conventional office materialized through furniture design, stretching from the metal framed desk with a wood top, to solid wood, and finally to a marble top for those at the top of the organization.⁵ Desks were not the only attribute of distinction; journalist Vance Packard described the importance of office size and location as well as all of the other accessories and elements of décor that were tied to status including: lounge furniture, ashtrays, filing trays, carafes, carpets, drapes, and even art. Within this elaborate system of office hierarchy, each change in organizational status merited new office décor such that nearly every promotion required a parallel change in office space.⁶

In their corporate office designs, postwar American architects and designers such as Eero Saarinen, Charles and Ray Eames, and Florence Knoll championed a total design approach, applying a modern design aesthetic in a consistent and predictable way as a spatial and material expression of organizational efficiency and modernity. For example, in her work leading the Knoll Planning Unit, designer Florence Knoll created meticulous plans for her office designs. Figure 4 shows an office interior featured in a Knoll catalog, showcasing the orderly arrangement of Knoll desks with their clean geometric forms providing an



Figure 4 Sales office for Deering, Milliken and Co. in New York City, designed by the Knoll Planning Unit, featuring Florence Knoll desks and Eero Saarinen desk chairs, 1958. The image illustrates the meticulous office planning that came to define the modern corporate interior.

Source: Image Courtesy Knoll Archive.

elegant contrast with Eero Saarinen's organic office chairs. According to Bobby Tigerman, Knoll's approach to design, which used the iconic pieces from the Knoll company's furniture line, became a "visual code" of the modern office.⁷ This style of corporate modernism became the common visual and material lexicon of organizational hierarchy and management through the midcentury in the United States.

Architects' and designers' emphasis on creating a rationalized and carefully planned corporate interior explicitly designed around organizational hierarchy was a material expression of an idealized and fetishized corporate bureaucracy.

In *White Collar*, C. Wright Mills describes the office as a “symbol factory” where every action and decision of the organization was rendered material through pieces of paper circulating through the office from one clerk to another, from one floor to the next.⁸ From the early-twentieth century through the postwar period, this circulation of paperwork was also a central focus in office design. Manuals on office layout and management emphasized the importance of arranging clerical staff around the idea of paperflow, ensuring that each worker’s placement fit with the circulation of paperwork (forms, memos, processes) snaking through the office from worker to worker and division to division.⁹

This design process awarded workers in the upper level positions with the elaborate perquisites of organizational status and treated low-level workers as operational gears in a bureaucratic machine. The workspace was an explicit manifestation of hierarchy, status and position. As Lange argues, this meant that “one’s job was visible with a glance at the purpose built desktop; its importance to the firm visible with a glance at its proximity to the powerful.”¹⁰ The postwar corporate office was thus conceptualized as a materialization of the working process, a tool in the circulation of work and information, and an outward expression of organizational hierarchy.

Cultures of Conformity and Discrimination

Workers were present in this system of circulating paper, but architects, designers, and managers often treated them as movable and interchangeable components within the larger space of the office and the organizational bureaucracy. This way of thinking about the organization as a system constructed predominantly of paperwork was not new or unique to the postwar era; for example in his research on the expansion of clerical work in mid-nineteenth century America, Michael Zakim analyzes the numerous ways in which the production and circulation of bureaucratic paperwork came to define modern capitalism, modern office work, and the design of the office itself.¹¹ Throughout the modern era, office design has been conceptualized by architects, designers, and managers as an idealized manifestation of the corporate hierarchy and more specifically as a system composed of workflows, job titles, and organizational relationships embodied by the corporate organizational chart. But the process of designing new corporate buildings in the mid-twentieth century with newly defined space standards created a fresh interpretation of the rationalized and standardized organizational hierarchy. Postwar architects imagined the corporate hierarchy, the organization, and its workers as abstract entities, and treated the new corporate office as the conceptual and material matrix that would draw these disparate components together and create a rational, cohesive, and orderly structure for

their containment. Interior layouts neutralized the human workers into movable and interchangeable widgets in the organizational system, and this approach to design reinforced an image of the workplace as one that was inhabited by a universal and illusory worker whose body, identity, and individuality had been scrubbed away.

As Aimi Hamraie has argued in their book *Building Access*, architects and designers have a long history of designing around a universalized figure whose “neutral body” is meant to stand in for an average and idealized user or inhabitant. Hamraie refers to this universal figure at the center of architecture and design practice as the “normate” user whose average body is not only used as an imaginary template for the design process, but also as a kind of prescription of the idealized user who was typically conceptualized as a White, male, and nondisabled user. In the postwar period, women were increasingly included in human factors charts and architectural planning, yet like their male counterpart, her idealized and average form was also imagined as White and nondisabled, again reinforcing this normate idea.¹² Hamraie’s analysis of the normate user is useful in thinking about the postwar corporate office because it offers a way of examining the assumptions of identity embedded in this idealized and imaginary worker at its center. Reading an office as a neutral arrangement of walls, partitions, furniture, and accessories without considering the real human bodies and their identities misses the ways in which these spaces reinforced and reproduced the entrenched systems of discrimination of the time.

In the postwar period, the imaginary worker at the center of the corporate office was the so-called organization man. Coined by William Whyte, the term referred to a typical White male worker in the upper levels of the organization, who had become fully enmeshed in the life and culture of the organization and who conformed with the social and cultural norms of the organization. Whyte describes the organization and its workers as one shaped by a culture of conformity that was cultivated by the organization not only in their day-to-day operation, but also through their hiring and promotion practices, which prioritized workers who would “fit in” and systematically weeded out workers who were in any way different from the status quo.¹³ In the 1950s and early 1960s, the American public was fixated on the problems of conformity not only in the modernist corporate office, where the sea of men in identical grey flannel suits became a literary and visual trope of the period, but also in the suburban landscape where the houses and families were similarly represented as expressions of the broader culture of visual and social conformity.¹⁴

The organization man was as much defined by the privileges of his race, gender, ability, and social class as he was by his organizational position. In this sense, the postwar corporate hierarchy, which served as the underpinning of conventional office design, was not merely a materialization of abstract jobs and

positions, but an inhabited landscape of privilege and opportunity for some, and exclusion and discrimination for others. As sociologist Joan Acker has argued, organizations are “inequality regimes,” shaped by systems of racism, sexism, and classism.¹⁵ Reading the postwar corporate office and its structures of hierarchy through the lens of discrimination highlights the ways in which these spaces were exclusionary by design. The oppression and discrimination of this period cut across numerous social identities, and manifest in many ways, but the burdens of discrimination were not carried equally by all affected by it.

One of the major divisions in American corporate offices of the postwar was along gender lines. White women had a significant presence in the office, yet their positions were generally constrained within their own internal hierarchy in which their highest position was often the role of executive secretary, or a managing or supervisory role in a clerical or support unit (generally comprised entirely of women).¹⁶ This was not new to the postwar period, indeed such sexism in office work dates back to the introduction of women in low level clerical roles in the late nineteenth and early twentieth century.¹⁷ Women’s work in corporate offices of the midcentury was usually divided between the highly visible front-office work (e.g., serving as a secretary), or back-office work in which workers were typically tied to certain machines and tasks. Women’s opportunities for advancement and promotion in this period were often shaped by their education, race, and social class such that highly educated middle class White women received the more plum and visually prominent roles like executive secretary while working-class women and women of color were often relegated to the back-office jobs.¹⁸ Women in clerical and secretarial roles were also frequently evaluated and promoted to higher paid and more visible positions based on their physical appearance (including dress, comportment, grooming, and perceived beauty). This meant that young attractive women were sometimes promoted over women who had more experience, and these systems tended to favor women whose appearance conformed to conventional standards of beauty at the time, which was deeply exclusionary particularly in terms of race, social class, age, body size, and ability.¹⁹

In addition to gender, American workplaces of this era were starkly divided along racial and ethnic lines. Racial and ethnic identities are themselves complex and intersectional, and different racial and ethnic groups in the United States experienced distinct types of racial oppression. During the final decades of Jim Crow in the United States, American cities, neighborhoods, schools, and workplaces remained structured by a landscape of legally authorized racial segregation that explicitly targeted Black Americans. As Patricia Hill Collins argues, although Black men and women achieved significant gains in civil rights through their activism (protests, sit-ins, marches, demonstrations, boycotts, and legal actions), even their legal successes, from *Brown v. the Board of Education* in 1954 to the passage of the Civil Rights Act in 1964, could not undo the historic

systems of discrimination, racism, and exclusion that were deeply embedded in American society.²⁰

Alongside Black Americans, Hispanic or Latina/o, American Indian, and Asian American workers also fought to gain greater educational and employment opportunities in the postwar United States with varying degrees of success.²¹ In their 1966 annual report on employment patterns in private industry, the Committee on Equal Employment found that compared to their White counterparts, Black, American Indian, and Hispanic or Latino men generally held the lowest paying jobs, and were severely under-represented in the highest paying job categories, particularly white-collar positions (professional, technical, and managerial jobs). Asian American men reportedly had somewhat higher participation in professional and technical positions than other workers of color, though they were also under-represented in managerial roles. Women of color in the United States had higher representation in clerical and service roles than their male counterparts, though they had fewer opportunities for professional, technical, or managerial roles than men of color. One notable exception being Asian American women who had somewhat higher rates of professional and technical employment than other groups, even when compared to White women at the time.²² Clerical divisions were therefore significantly more racially diverse than other white-collar job categories, but the overwhelming majority of office workers in this era were White. This racial landscape only became more pronounced as many large companies began to move their offices out of the urban centers where people of color were often concentrated and into suburban areas that lacked public transit thereby further deepening the entrenched racial segregation of white-collar work at the time.²³

Workers with disabilities also encountered significant barriers to jobs and opportunities in this era including outright discrimination as well as a lack of accessibility. Disability activists and organizations fought for greater inclusion and opportunity in the postwar, but opportunities for disabled workers remained limited.²⁴ As Bess Williamson describes in her book *Accessible America*, veterans injured in the war had some government support geared toward rehabilitation and the goal of a productive career following their military service, but civilians with disabilities had far fewer resources or support in navigating housing, jobs, and transportation. As Williamson argues, for both veterans and civilians, access was often further restricted based on race, social class, and gender.²⁵ At the same time, the lack of architectural standards to support access in the mid-twentieth century meant that people with disabilities were not only excluded from certain types of jobs (and the educational opportunities to get those jobs), but in some instances they could not even enter the building.

Beyond gender, race, and ability, workers also experienced discrimination on other aspects of identity such as social class, nationality, education, religion, and sexuality, among others. These various classes of identity intersect and interact

with each other such that workers who identified with marginalized identities in multiple categories experienced deeper degrees of discrimination.²⁶ Returning to our “organization man” that has so defined discussions of postwar offices and postwar corporate culture, we have to recognize the numerous workers whose bodies and identities were expressly excluded from those jobs and their assigned spaces. Further, the presumption of this “organization man” and his normate body was also physically inscribed into the design of the office from the ableist architectural spaces to the sexist office chairs and ambient air temperatures, creating spaces that were optimal for some classes of users, and ill-suited or exclusionary for others.²⁷ The rigid corporate hierarchies of the postwar, so carefully reproduced in office design and office space, are inseparable from the culture of exclusion, including entrenched racism, sexism, and ableism that served as the understructure of organizational culture.

Corporate Countercultures

In the 1960s, an emerging youth culture challenged these entrenched cultures of conformity, discrimination, and technocratic bureaucracy that had defined postwar culture. The activism associated with the various civil rights movements, the women’s movement, the peace movement, and the new counterculture left a significant imprint on American business. The civil rights, disability rights, and women’s movements paved the way for increased representation for more diverse workers in a greater array of white-collar jobs, increasing access and opportunities for some women, workers of color, and workers with disabilities, but White able-bodied cisgender men continued to dominate professional, technical, and leadership roles across most industries and in most American organizations.²⁸ At the same time, corporate advertising of the late 1960s and 1970s increasingly adopted the aesthetics of the counterculture through psychedelic imagery and youthful slogans. They also echoed some of the social activism of the era for example selling their products with the language of women’s liberation and Black power even as they failed to meaningfully engage in structural change to increase the diversity of their organizations.²⁹

As corporations were transforming their external image to reflect the new counterculture and attract a youthful and more diverse consumer market, many were also refashioning their internal corporate culture throwing off the scourge of top-down bureaucracy in favor of a new and more progressive approach to management and organizational structure. Some corporations began to see their own corporate culture as a hindrance to economic growth. In other words, after years of carefully nurturing and developing an army of “yes men” in grey flannel suits, some organizations began seeking ways to upend those social norms and infuse new energy, fresh ideas, and even construct an entirely new

corporate image and corporate culture. At the time, new theories of management, organizational culture, and human motivation were actively challenging many of the assumptions of conventional postwar corporate culture.³⁰

Among the most influential of these new management theories was the work of Douglas McGregor. In 1957, McGregor published an article titled “The Human Side of Enterprise,” which he expanded into a book published in 1960 of the same title that described two competing theories of management: Theory X and Theory Y. The “Theory X” approach was characterized by several key assumptions about the average worker, who: wants to avoid work; dislikes responsibility; is indifferent to the needs of the organization as a whole; and must be controlled by management through close supervision and micromanagement of all work-related tasks and behaviors. In contrast, so-called “Theory Y” managers believed that the role of management was not to “control” workers, but to foster a more supportive workplace environment in which workers had greater ownership for the work process, greater responsibility in the organization, and more opportunity for development and growth.³¹ Through the 1960s, “Theory X” was often framed as a “conventional” management approach characteristic of an old-fashioned corporate bureaucracy, and “Theory Y” was typically heralded as a more progressive approach appropriate for a new age of organizational culture and a new generation of workers.

As the concept of “Theory Y” was gaining ground among forward-thinking managers and administrators, management consultant and theorist Peter Drucker’s concept of the “knowledge worker” was also enormously influential. First coined by Drucker in 1959, the term “knowledge worker” was frequently used in the 1960s to describe a new generation of professional workers who were more independent than previous generations of office workers. In contrast to the corporate drone of yore, the “knowledge worker” was characterized by Drucker as a highly educated expert whose work required thinking, collaboration, originality, and creativity rather than paper pushing. Most importantly, Drucker argued that the “knowledge worker” was outside of conventional organizational hierarchies. This new class of worker operated independently, made decisions, and ultimately had more in common with executives than with managers or technical workers, and consequently required managers to adopt a new approach that fostered each worker’s need for autonomy and individuality.³²

These ideas were increasingly popular in business circles through the 1960s and early 1970s. In his bestselling book *Up the Organization*, businessman Robert Townsend described his personal philosophy of management and business as president and chairman of rental car company Avis. In many ways, Townsend embodies the countercultural business approach through his management style, and he specifically cites Douglas McGregor’s *Human Side of Enterprise* and Peter Drucker’s *Managing for Results* as two of the most important books on management he had ever read. In other words, for Townsend, McGregor and

Drucker were his twin pillars of influence in terms of his overall management and leadership style.³³

Organizations viewed the adoption of these new theories of management as strategically important because research at the time suggested that young workers in the late 1960s and early 1970s had very different motivational goals than their parents. A national survey conducted from 1968 to 1971 found that a growing number of young people sought work that would be personally fulfilling or “meaningful.” The study reported that these young workers were resistant to conventional hierarchies, instead prioritizing jobs that would give them greater autonomy to make decisions and independence in the workplace. As a 1973 US government report *Work in America* described, young workers were not revolting against work, but rather revolting against a culture of “authoritarianism.”³⁴ This idea, that young workers were a new and different class requiring a new organizational culture, led some businesses to rethink their organization to better appeal to them. For organizations, reducing hierarchy through the adoption of new management theory was a strategic change that would ultimately help them attract and keep younger workers.

Together, “Theory Y” and “knowledge work” represented a shift in thinking about the nature, purpose, and structure of organizational hierarchy in the 1960s. Rather than conceptualizing the corporate structure as an army in which all of the workers were marching in bureaucratic lockstep, advocates of this new kind of corporate structure characterized it as more fluid, dynamic, and “organic” in its form. Drawing on the ideas of Ludwig von Bertalanffy’s general systems theory and Norbert Wiener’s concept of cybernetics, organizational theory increasingly described organizations as complex and interdependent systems.³⁵ As Reinhold Martin has described, this emerging idea of the organic system in the 1960s challenged the centralized and top-down organizational ideal with a new decentralized and distributed model of the organization, which operated “without hierarchy or center.”³⁶ All of these new ideas challenged conventional notions of corporate hierarchy and required a significant philosophical shift in organizational culture. For advocates of this new way of thinking about management and organizational structure, upending hierarchical practices was not a simple or straightforward process that would transform the organization overnight, but rather an ongoing and evolving set of interactions among workers at all levels, and these new ideas were vital to the development of the new open plan office.

Designing the Egalitarian Office

In the late 1950s, just as Douglas McGregor was first writing about his concept of Theory Y, and Peter Drucker was just introducing the concept of the knowledge

worker, German organizational consultants (and brothers) Eberhard Schnelle and Wolfgang Schnelle began experimenting with a new way of designing the office that they called *burolandschaft* or “office landscape.” According to Andreas Rumpfhuber, theories of cybernetics inspired the Schnelle brothers leading them to conceptualize the office as a dynamic system of information and people in constant circulation. One of their first experiments in office planning was at a publishing house in Germany in the late 1950s, where the Schnelle brothers used the wide expanse of a warehouse space to reprogram the organization around internal patterns of communication. Rather than breaking up the space with enclosed offices, the Schnelle brothers arranged workers in the open warehouse in loose groupings of simple desks, mixed with a modest number of low movable partitions and plants.³⁷

Soon after these early experiments, the Schnelle brothers moved their offices to the Quickborn district of Hamburg, renamed their company the Quickborner Team, and began consulting on office projects across West Germany and eventually throughout Europe. Early reporting from the mid-1960s on the office landscape concept in the United States often treated the idea as an office planning oddity from Europe, but in the late 1960s office landscaping started to get some attention, most prominently from Carroll Cihlar, editor of the journal *Office Design*, who published the first extended article on the concept in March of 1967 and organized a symposium on the new concept held in September 1967.³⁸ At that same time, Quickborner was just opening their first American office in Millburn, New Jersey headed by Hans Lorenzen and Dieter Jaeger, and installing their first American office landscape in the offices of the Freon division of DuPont, where a single floor in a conventional building was redesigned using the new office concept (Figure 5).³⁹ It is unclear why Quickborner chose this moment to open a US office, but based on their promotion of the concept in the American press at the time, Quickborner perhaps saw a window of opportunity for the landscaping idea as these new ideas about management theory were gaining ground in American architectural and business circles.⁴⁰ Following the highly publicized DuPont installation, the American branch of the Quickborner Team was soon working with a number of organizations around the United States including John Hancock Insurance, Eastman Kodak, Port Authority of New York, and Purdue University.⁴¹

As the Schnelle brothers were initially developing their office landscaping concept in the late 1950s, American designer Robert Propst was also becoming interested in the problems of office work and office design. A design consultant, Propst joined Herman Miller as the head of their Research division in 1960, and promptly started work on the problem of office furniture and office design. The result of his early experimentation, research, and thinking about the needs of workers and the problems of modern work was a new kind of furniture called Action Office.⁴² The first iteration of Action Office, released in 1965, was a

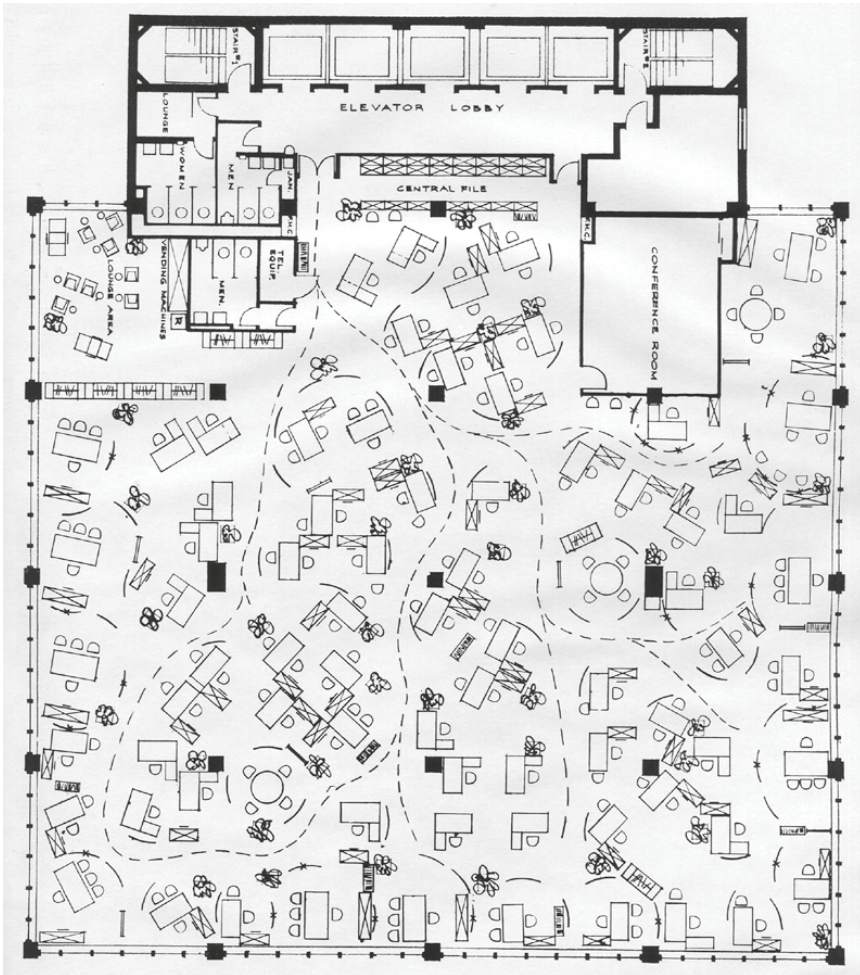


Figure 5 DuPont Freon Division layout, designed by the Quickborner Team in 1967. The design is characteristic of the office landscape layout with its loose arrangement of desks and curving screens creating modest privacy.

Source: *Office Landscaping: An Open Plan Concept of Office Design*, 71. Image used with permission of Combine Consulting.

series of elegant wood and metal office furniture pieces that were designed to support the unique needs of executives in the office, particularly their need for manipulating information, supporting natural clutter (as a manifestation of the thinking process), and their need for physical movement within their workspace. Action Office 1 was not a market success, but Robert Propst and his research team soon began work on an expanded version of Action Office that would take some of the same ideals embedded in the original series into a modular furniture

system suitable for everyone in the office. At the time, Propst was aware of the work of the Quickborner Team and their office landscaping concept, which was just starting to gain significant attention in the United States. In May of 1967 he attended a conference on office landscaping in Stockholm and spoke at the symposium later that same year on office landscaping organized by the American journal *Office Design*. For Propst, the office landscape concept represented an exciting new direction in office design that reinforced much of his own thinking about the future of work.⁴³

In 1968, Herman Miller released Action Office 2, a novel furniture system featuring a series of low freestanding partitions that could support work surfaces and storage components. As a system, Action Office offered a suite of interchangeable and interlocking components that could be assembled in many configurations and combinations to meet the individual working needs of any worker in the office. Figure 6 shows an early installation of the new Action Office furniture system in the Citizens and Southern National Bank in Atlanta from 1968. Like office landscaping, Action Office imagined the workplace as an open interior defined through the careful placement of modular furniture components rather than defined by walls. In tandem with the release of the new modular furniture system, Robert Propst published a kind of manifesto called *Action*



Figure 6 Image of Action Office 2 installation at Citizens and Southern National Bank in Atlanta, Georgia, 1968. In the image, the modular partitions and storage of Action Office serve to define the individual workstations, while still maintaining an open interior.

Source: Courtesy of Herman Miller Archives.

Office: A Facility Based on Change, which described the furniture concept and its underlying philosophy.

The coincidental timing of the release of Action Office 2 just as office landscaping was taking off in the American architecture and design press meant that the articles introducing the office landscaping concept to an American audience often referenced the new Herman Miller furniture as a related and even complementary idea.⁴⁴ Despite their technical differences, Action Office and office landscaping shared some common ideas including a belief in the need to support the new developmental management approaches embodied by the work of Douglas McGregor and Peter Drucker, and a desire to reduce the role of hierarchy in office design.⁴⁵

The American architecture and design press described these new open plan office concepts as naturally egalitarian. For example, a 1969 article in *Office Design* described the newly landscaped offices of a jewelry company in Rhode Island as “democratic” because of its minimal distinction among workers in the office. According to the article, the furniture used for the executive offices was the same as that for the “humblest office boy’s work station.” The most elaborate private office was not for the president, but for the personnel department who required it for interview purposes. Describing their broader intention, the Research and Design Institute, an architecture and design office responsible for the project, described how the new plan was meant to “provide superior working conditions” and an expression of a “progressive” business culture. According to a published statement from one of the designers, the new building “is organized on a democratic basis, expressed in office landscape, which supports individual initiative, team effort and a feeling of human dignity for the occupants.”⁴⁶ Throughout both the article and designer statement, office landscaping is treated as a natural expression of this new culture of work that upended conventional ideas about organizational hierarchy and status in office planning. By rejecting private offices and sharply reducing the gradations of hierarchy in office design as well as their symbolic indicators, advocates of these new open plan concepts promised to reimagine the office as a more egalitarian space in which the markers of hierarchy were all but eliminated.

This rejection of hierarchy embedded in early open plan designs manifest primarily through the elimination of private offices. Of course, advocates of the open plan recognized that some workers were reluctant to give up their private offices. For example, *Progressive Architecture* described how the more established workers in upper positions may not mind leaving their “prestigious den” for an open plan design but the “lesser fry” of the office who were just able to get a private office in the conventional system were likely to feel “deprived of the door and the rug that have been their symbols of success and the tangible goals of their striving.”⁴⁷ In other words, according to the article, those whose status was clear, unambiguous, and well established would be able to accept

this change in space, but those just on the margins of earning a private office were most likely to balk at the loss. This typical framing reinforced the belief among advocates of the open plan that the private office was not a functional requirement or necessity for these upper level workers, but rather an empty status symbol and even a barrier to effective communication and interaction.

The aesthetics of these early open plan concepts also rendered the organizational structure and its hierarchy almost illegible by its design. In contrast with conventional office planning which followed the form of the organizational chart, new open plan offices had a visually chaotic appearance in which the myriad divisions and areas appear to merge and flow into each other. This deliberately disordered quality of the open plan office reduced the spatial prominence of organizational hierarchy. Although hierarchy was still present and visible, the differences of status in these early open plan designs were subtler than in conventional office design, typically conveyed through individual floor space, the adjacent number of partitions and plants, and the specified furniture. As environmental psychologist Franklin Becker argued, the increased use of the new class of systems furniture such as Action Office provided a common design vocabulary across the space in open plan offices that often overpowered the differences such that the various workspaces looked more similar than different.⁴⁸ Simply by eliding the familiar signifiers of status, the open plan challenged the conventional uses of hierarchy in office design, and by extension its emphasis on organizational culture. According to architect Frank Duffy, in these early types of open plan or office landscaping designs “egalitarianism was worn conspicuously as a badge of progress” symbolizing an office’s democratic culture and their flat organizational hierarchy.⁴⁹

As the open plan became popular, advocates of the concept viewed its egalitarian associations as naturally youthful and even inherently countercultural by design. When McDonald’s chose an open plan office for its corporate headquarters completed in 1971 in Oak Brook, Illinois, descriptions of their office in the press characterized their choice of an open plan as a reflection of their progressive corporate culture. In news stories, reporters even described the McDonald’s office as youthful, with rock music playing in the cafeteria and workers dressed in fashionable “mod” clothing.⁵⁰ For McDonald’s the youthful organizational culture was an important strategy for attracting and keeping young talented workers. In the words of one of the executives at McDonald’s, “if you are going to hire young people ... you’ve got to motivate them and these days little cubicles and superficial status symbols like northeast corner offices just don’t motivate people’.”⁵¹ The new offices featured interiors designed by Associated Space Design (ASD), an Atlanta interiors firm led by William Pulgram. They adopted a new furniture system originally created for the McDonald’s offices called TRM or Task Response Module, which was designed by Pulgram and manufactured by Eppinger.⁵² The TRM system (Figure 7), with its natural



Figure 7 Eppinger's TRM workstations in the McDonald's Office, 1971. ASD arranged the elegant wood furniture to create individual professional workstations with integrated partitions and storage components.

Source: ASD|SKY.

wood grain and case furniture aesthetics, was expressly designed to appeal to executives by providing a more sophisticated appearance relative to the other classes of systems furniture available at the time.⁵³ As an article in *Interiors* described, the “wall free corporate democracy” of McDonald's offers a space where “every member of the work force enjoys high-echelon amenities, views, and space.” As many of the stories emphasized, even founder Ray Kroc worked in one of the TRM open plan workspaces. Yet, the supposedly “status-free space” also maintained clear distinctions of status in terms of the allocation of floor space, the selection of furniture, and the height and degree of enclosure afforded by the furniture; secretaries and clerical staff, for example, worked at regular Steelcase 5200 line desks.⁵⁴

Perhaps nothing embodied this countercultural aspect of the McDonald's open plan corporate office more prominently than the “think tank” space—a futuristic padded spaceship-like structure on the eighth floor of the building. Prefiguring the playful and immersive office designs associated with Silicon Valley of the 1990s and early 2000s, the McDonald's think tank was designed to provide workers privacy for thinking and escape during the work day. The space had a labyrinth leading to a work room with beanbag chairs and a kind of hatch opening into a round “meditation room” featuring a circular 700-gallon waterbed, fully enclosed with curving walls, and containing a stereo system



Figure 8 Image of a man sitting cross legged on a waterbed in the round meditation room inside the McDonald's "Think Tank," 1971. The think tank offered workers a retreat from the office that would support creative thinking.

Source: ASD|SKY.

and a projection system for images. Available to anyone in the company by appointment, the think tank gave workers a place to unwind, reflect, and escape from office distractions (Figure 8). The space was used both by individuals and small groups (up to six workers) for meeting and brainstorming, and it was even reportedly used for some personnel interviews. Just imagine the strange and perhaps unnerving experience of a job candidate showing up for an interview at the McDonald's headquarters and finding themselves escorted into the think tank room where they might be expected to crawl through a hatch and plop down on a waterbed. While such a scenario might have been a bit odd for a male candidate applying for a job, the same situation for a female job candidate would have likely felt much more unsettling because of the sexual implications of being interviewed on a waterbed; imagine the awkwardness of sitting on an undulating waterbed in a skirt and tights. There is even some suggestion in the reporting that McDonald's "think tank" space was used for sexual dalliances between co-workers.⁵⁵ In this sense, the "think tank" waterbed in McDonald's open plan office implicitly referenced the sexual revolution of the 1960s and provided a prominent symbol of McDonald's youthful corporate culture, which in turn reinforced the countercultural aspects of their open plan office. The framing of the open plan office as not only egalitarian but also inherently youthful

and countercultural meant that the open plan itself was often seen as a spatial expression of this new type of organizational culture.

The Politics of Planning

In August 1972, the Bureau of Planning and Budget in the State of Wisconsin requested a remodel of their office space in the Hill Farms building in Madison, Wisconsin. The original Hill Farms office space, completed in 1964, had utilized full-height movable partitions for private offices, but the Bureau of Planning and Budget asked for those old partitions to be removed entirely to implement a new “office landscape” with no private offices at all. In the memo, the assistant director of the office cited the numerous benefits of the office landscaping approach, particularly its relationship to the “management philosophy” of the planning office and the “developmental nature” of planning program’s work. The memo indicated that the staff was so enthusiastic about this proposed change that if not approved it would be a significant blow to their morale.⁵⁶

At around the same time (in February of 1973) another division in the State of Wisconsin, the Board of Vocational, Technical & Adult Education in the very same building had a “delicate internal management problem.” Their recently remodeled offices were reportedly quite lovely with “judicious uses of color and plant material to create a pleasing environment,” but according to an internal memo there were “dimensional worms in Eden.” One of the bureau directors had a private office that was 6 linear feet (182 cm) larger than the others because of the unfortunate placement of a radiator which was too expensive to move, and the other directors were very upset about this perceived inequity in office sizes. The original memo from the agency head to the Department of Administration (DOA) asked for permission to spend the additional funds to move the radiator and wall allowing the director offices to be the same size. In their response, the head of the DOA responded, “We are not unaware of the degree to which space layout matters tend to take on a seriousness of concern disproportionate to their relative importance in the overall responsibility of governmental management.” He continued stating, “we will proceed with the changes you have requested, but in all honesty with a general feeling of reluctance in so doing. Sorrow might even be a better word—for, in this instance, those of us who are attempting to fight the war of negativism about bureaucracy have lost a small battle in that process.”⁵⁷

As one group of workers in the State of Wisconsin was enthusiastically eliminating their private offices in favor of a new open plan design, another group was struggling with infighting over the equitable representation of hierarchy in terms of office size. Together, these two stories reflect the emerging debate of this period about the meanings and uses of hierarchy in office design as the

open plan became mainstream in the United States. Like many organizations around the country, in the early 1970s the State of Wisconsin had already begun to recognize the burden of bureaucracy and cost associated with the obsessive fixation on status in office design associated with conventional planning, and the new open plan was increasingly attractive to organizations like Wisconsin State Government in part because it upended those conventions. By the late 1970s, the State of Wisconsin would eventually adopt the open plan concept as a standard for all state government office buildings.⁵⁸ Yet, even as it became a state standard in Wisconsin, a number of individual workers and even entire divisions remained reluctant to accept the open plan and the parallel loss in space and status that it represented.⁵⁹ The leadership of the DOA in the State and some units and their workers readily adopted the open plan concept in the late 1970s and early 1980s, but there was also significant internal resistance at all levels and across numerous units, and the resistance was constantly being negotiated through internal debates about the relative allocation of space for divisions and their workers and the specification of furniture.⁶⁰

These kinds of organizational tensions over office design were not unique to the State of Wisconsin. As noted above, early discussions about the open plan often referenced the potential angst and dissatisfaction of workers who would typically receive a private office. Even as the open plan became mainstream through the 1970s, concerns remained about worker resistance to the new design particularly with regards to the representation of hierarchy and status. According to their 1978 study, Steelcase and Louis Harris and Associates found that 88 percent of business executives surveyed acknowledged that they viewed the loss of status associated with the open plan as a drawback of the design concept.⁶¹ In the business journal *Administrative Management*, an article titled "Getting the Holdouts to Accept the Open Plan" describes how some workers experienced significant apprehension about the new design concept, and that some organizations continue to encounter "resistance and even hostility" from workers, particularly low level and mid-level executives "who may regard their private offices as 'status symbols'."⁶²

In part to address these kinds of concerns, office furniture manufacturers marketed systems furniture lines in ways that showcased its ability to express status and position through the specification of components and finishes that offered a visual language of hierarchy even in the open plan. For example, systems furniture price lists of the 1970s and 1980s provided increasingly elaborate gradations of finishes, textiles, and other materials creating an index of status embedded in the furniture specification process. From the laminate finishes to wood grain, from basic nylon to natural fibers or even leather, and from plastic accents to gleaming metal, the language of materials was not just one of aesthetics but also of relative cost and perceived value. Nearly every furniture system could be specified with different materials along with different

components throughout the space to ensure a clear and explicit representation of hierarchy. In addition to using the furniture as an expression of status, some projects also maintained a clear spatial separation between upper level executives and general office workers. At the new Montgomery Ward building completed in 1973, designed by architect Minoru Yamasaki with open plan interiors by Rodgers and Associates, top executives had an exclusive space on the twenty-sixth floor of the building. Although the twenty-sixth floor was also an open plan in its design, it featured an elegant wood furniture system that was also different in style and form from the standard Steelcase 9000 furniture used in the rest of the building.⁶³

Despite concern about the plight of the professional or executive worker in the open plan and their perceived loss of status, there was comparatively little concern about general office workers, particularly low-level clerical and support staff. In fact, most writings characterized the open plan as a net benefit for low-level workers because it was more egalitarian, more humane, and more supportive of work than the conventional office design.⁶⁴ For the lower-level workers in clerical and support roles (who again were predominantly women), the biggest status issue in office design was not the aesthetics and intricacies of status, but rather the degree of inclusion in the planning and design process.

Early advocates of the open plan often championed the idea of participation as an important attribute of the new concept. In office landscaping, the Quickborner Team surveyed all workers in an office-wide communication assessment that tracked the real interactions among workers and liked to include workers at all levels on the planning team. Similarly, Herman Miller provided its clients a detailed survey template to assess the individual working and spatial needs for all workers as part of the planning process. In his 1978 handbook, *Open Office Planning*, designer and writer John Pile cautioned architects and designers that in the design process for an open plan office both of these models could produce an overwhelming amount of data that in the end might not be very usable. Pile further states that such detailed surveys of low-level clerical and support staff are not useful because those workers are essentially “interchangeable units whose personal requirements do not differ in any important way from person to person.” Because this class of employees often had a high turnover rate, he argues, “efforts to identify personal needs and work patterns are often pointless.” For these reasons, Pile instructed designers and architects that there was little value in collecting such detailed information from all staff. Yet, Pile also acknowledges that involving workers could help encourage user acceptance.⁶⁵ How do we square this seeming contradiction?

When prominent advocates and practitioners of open planning like John Pile describe the benefits of user participation, they often meant upper level professional workers such as managers and executives, not low-level support staff. Architects and designers in the 1970s and 1980s increasingly included

upper-level workers in office planning, in some cases even giving those workers significant choice regarding their individual preferences. The Union Carbide office in Danbury, Connecticut designed by architect Kevin Roche in the late 1970s and completed in 1982 has often been characterized as an example of high user participation.⁶⁶ As part of the planning process for that project, Roche and his team conducted hundreds of interviews in the early planning stages with various (overwhelmingly male) executives and managers and most of those workers were given the opportunity to choose their own furniture from some thirty different options for their private offices. The mostly female clerical staff in the same building were only given minimal opportunities to participate in the planning and design. They were invited to visit a mock-up of the proposed furniture and fill out a simple survey evaluating the selected furniture (a fairly standard practice at the time), but the basic layout and open plan workstation design had already been more or less decided by the architect and corporate leadership.⁶⁷ Unlike their professional colleagues whose opinions and needs were carefully assessed and who were even allowed to choose their own office furniture, the clerical staff were given no real choices and few opportunities to participate. The entrenched social inequities of the workforce which were mapped into the allocation and design of space were also reflected in the design process itself through participation (or lack thereof).

The available evidence of the 1970s and 1980s suggests that in most office design projects, general office workers were not included in the design process much at all. Even when workers were involved it was in a very limited capacity, sometimes providing data in an initial survey, and sometimes providing comments on specific aspects of a workstation mockup, as in the Union Carbide example. The 1978 Steelcase survey found that the majority of general office workers had minimal participation in the design process, and even a majority of architects and designers surveyed acknowledged that they felt the users (general office workers) were not included sufficiently in the process.⁶⁸ Yet, post occupancy studies on participation at the time had shown a strong correlation between real participation in the design process and satisfaction with the completed building. In 1972, the General Services Association sponsored an experiment in the Federal Aviation Administration (FAA) offices in Seattle where the employees designed their own workstations. While the overall design was reportedly “very heterogeneous,” a phrase that suggests a lack of visual cohesion from a design perspective, one year later, after the building was occupied, the workers in that building were generally more satisfied with their space than were the workers in a conventionally planned FAA office in Los Angeles.⁶⁹ Perhaps not surprisingly, design manuals of the period encouraged architects and designers to create opportunities for workers to participate in the planning process in some way, but the inclusion of workers in the process was often framed as an effort to encourage their acceptance of the new office rather than as an opportunity to

receive meaningful feedback on their needs and preferences.⁷⁰ In other words, inviting worker participation was not even necessarily about gaining important insights from the worker to improve or otherwise change the design concept, but rather strategically deployed as a means of ensuring that the workers felt heard even if their ideas or comments were not reflected in the final design at all. Even when general office workers were included, they were rarely told or able to see how their input in the design process was used or addressed in the final design, which often only made workers more frustrated that their feedback, though solicited, was not taken seriously.⁷¹

This suggests the ways in which hierarchy, power, and status were not just reflected in the physical arrangement of the office, the combination of components, or the choice of materials used in a particular workspace, but it was also produced through the design process itself. Every office design was a negotiation between the organization, its space planners (architects, designers, facilities managers, etc.), its various divisions or units, and its workers across the organization. Those who had a voice in the process had an opportunity to shape the outcome to their benefit, or at least air grievances as part of planning, but for every worker included in the design process, there were often hundreds and even thousands of workers that were not included at all. Because corporate hierarchy remained shaped by discrimination and exclusion, participation in the design was also tied to identity such that White cisgendered and able-bodied men in leadership roles continued to have more voice in the process than women, workers of color, disabled and other marginalized or disenfranchised workers.

Exclusion by Design

When architects and designers first promoted the open plan in the United States, they saw this new office design idea as a transformation of the office and a reflection of a larger shift in thinking about the structure and culture of organizations. Some architects, designers, and organizations saw the conventional office with its “total design” aesthetic and carefully delineated corporate hierarchies as out of step with the new culture of office work in which the trappings of status were no longer useful or even relevant. For its advocates, the open plan was an expression of these new theories of management and organizational culture. Echoing the countercultural idealism of the late 1960s, advocates of the open plan in the United States saw this radical approach to office design as an almost utopian office concept that would promote a more egalitarian organizational culture through its design.

Champions of the open plan often characterized this egalitarian ideal as a transformation of the organization, but the idealized vision of the future of work was one that privileged and prioritized the needs, concerns, and priorities of upper

level workers. Despite their gratuitous nods toward supporting the file clerk in the open plan, architects' and designers' idealized open plan was never a place in which all workers and all positions were treated or even imagined as equal. This gap between the language of egalitarianism and its practical application manifest in the design of the open plan, which often continued to silo top level executives from other classes of workers, and also in the planning process in which low-level workers including clerical and service workers were rarely given the opportunity to meaningfully participate in the design process. The hierarchical aspects of the open plan office persisted both materially and conceptually through the design process, which reinforced and reproduced organizational privilege. As organizational theorists Karen Dale and Gibson Burrell write, architects were mediators between the building and their corporate clients, and the resulting organizational architecture thereby reproduces and reinforces a "web of power relations" within the organization. According to Dale and Burrell, this reification of power, status, and hierarchy in architecture is an artifact of these deeply rooted power relations that are endemic to the architectural process itself.⁷²

Office design has historically inscribed hierarchy and power into space, but through their use, offices are inscribed with human bodies whose identities also shape the meaning and use of that space. The space of the office, though technically planned as a system of interlocking workspaces designed for abstract positions, was always eventually inhabited by workers whose bodies and identities were as much a part of the space and the organization as their jobs. The legacy of discriminatory hiring and promotion practices in American organizations produced a landscape of inequality and exclusion based on markers of identity including gender, race, and ability. Despite its egalitarian idealism, the open plan was never a neutral description of an abstract organizational entity, but rather a materialization of the systems and structures of power that were themselves shaped by a long history of discrimination.

Chapter 2

Managing Change

For advocates of the open plan, the rejection of organizational hierarchy and the embrace of new management theory was an essential underpinning of the concept, but in selling the open plan to clients, the most frequently referenced benefit was the open plan's flexibility to support change. Architects and designers argued that the office should be as nimble, flexible, and adaptable to change as the organization itself, and offices should ideally reflect and support organizational change. In office landscaping, the Quickborner Team deliberately favored light simple furniture and movable pieces (in some cases on wheels) that would allow the entire space and its contents to easily and quickly adjust to changing organizational needs. Similarly, Robert Propst and his team designed Action Office as a highly modular system of components that would allow organizations to make quick changes to their entire layout to support ongoing change. Proponents of the open plan imagined a future in which office change was constant and immediate; the office and the organization were locked together in perpetual synchronicity—the one immediately changing in response to the other. In their fixation on the importance of change, advocates of the open plan conceptualized the office as an ever-evolving and changing space that was nimble and responsive to every organizational, technological, and social change.

To think about change is inevitably to consider time as a dimension and factor in the use of space. Of course, office design itself exists in time—every office space reflects the assumptions and needs of its particular moment, and those assumptions are reproduced materially and spatially. Office design is also a process that takes time, often requiring many years of planning and discussion before the builders place the first stone in the foundation. The open plan imagined the office not as a sealed time capsule that would exist unchanging for decades, but rather as a dynamic system that would continue to change and evolve in the days, weeks, months, and years following the initial installation. Advocates of the open plan viewed change as a practical necessity in office design, and as an organizational value that reflected the progressive management ideals of its time.

Yet, the meaning and purpose of organizational change evolved significantly from the late 1960s when advocates first began heavily promoting the open plan in the United States to the 1970s when the open plan and systems furniture became mainstream in American offices. Though born in an era of economic prosperity that treated change as an expression of growth and innovation, by the 1970s, the changes underway were markedly different. A significant economic downturn, rising inflation, a loss of productivity, and multiple energy crises helped create a deeply unstable economic climate. Change in the 1960s was framed as optimistic and forward-thinking, but change in the 1970s was often implemented by organizations in a reactionary and defensive manner, frequently with the express goal of increasing productivity, decreasing energy usage, reducing space usage, and cutting operational expenses. The design, evolution, and use of the open plan reflect both the idealism of its early development as well as the pragmatic concerns of organizations that adopted the open plan in large numbers through the 1970s and 1980s.

While the open plan remained an expression of change, the meaning and purpose of organizational change in the 1970s was altogether different from the idealistic vision of change so often described in the 1960s. As I argue, this gap between the original vision and the reality of implementation was materialized in the design of systems furniture which was conceptualized and promoted as a tool for easy change, but was by design a complex system of components that only became more cumbersome as it grew and changed to address emerging technical needs. For organizations, managing office changes required a facilities management team to maintain the ever-growing inventory of systems furniture and to plan, manage, and implement ongoing organizational, technological, and spatial change. For workers, this constant change manifest in significant technological, social, economic, and organizational upheaval that produced an increasingly unstable environment in which workers jobs or units might cut, create, or change entire units overnight. The perpetually shifting partitions of the open plan were a spatial manifestation of the growing economic and social instability.

Designed for Change

The idea that change was an important attribute of office design was not really a new idea in the 1960s. From the early twentieth century through the postwar period a significant number of American corporate offices, factories, schools, and hospitals were designed with full-height movable partitions. Among the major manufacturers of full-height movable partitions was the E. F. Hauserman Company founded in 1913 in Cleveland, Ohio. The Hauserman Company marketed their partitions as a valuable amenity that would afford organizations the

flexibility to adapt their interior space to inevitable changes over time. Integrating interior raceways for running electrical wiring and telephone lines inside the panel, and providing hardware for doors, windows, and other architectural details, the partitions functioned like regular interior walls with the added benefit of movability allowing organizations to more easily adapt the space to their ever-changing needs. In their marketing materials from the 1930s, Hauserman described how their steel walls could be rearranged again and again making them “permanent assets” that would actually “outlive the building.”¹

By the postwar period, Hauserman boasted a sophisticated range of partition types suitable for every kind of space from the executive suite to the dispensary and featuring a range of materials, colors, and options to provide architects and organizations a robust and flexible system suitable for specification throughout a building. Hauserman described their movable walls as prefabricated and modular by design, featuring standard posts and components that allowed easy and efficient assembly and disassembly. One brochure from the 1950s even depicts the walls being changed in just thirty minutes as a secretary and executive continue their work, apparently undisturbed by the team of Hauserman workmen dismantling and reassembling the wall system in their midst.² In fact, even as they celebrated the ease with which their partitions could move, Hauserman also emphasized the value, importance, and even necessity of using Hauserman’s own installers not only for the original installation but for any subsequent move. In buying Hauserman partitions, the organization was not just buying the partition hardware, they were also purchasing Hauserman’s installation services for its lifetime of usage.³ They celebrated their nation-wide system of representatives and installers, and the Hauserman logo prominently featured a workman holding up a panel in the shape of the letter “H”, reinforcing the service orientation of the company.

Architectural historians and theorists have often characterized the prominent use of movable walls in postwar architecture as part of a modular understanding of architectural space. In his book *The Organizational Complex*, architectural historian Reinhold Martin describes how postwar corporate architecture embraced standardized modules as key components of architectural design. Martin argues that the concept of the “module” served to structure office space in standardized or prefabricated components within the architectural envelope allowing for some flexibility but constrained by the rigidity of modular standardization. While ostensibly designed to enable change, that change was circumscribed as a system of fixed components that could only be arranged in a limited set of configurations.⁴ According to Martin, this emphasis on modular systems was tied to the use of grids as the architectural ideal, embodied visually by the curtain wall with its glittering grid of glass.⁵ Creating structures of “organized flexibility” allowed architects to use standardized building components as an architectural language made up of pre-fabricated assemblies, modules, and parts.⁶

New York City's Union Carbide building designed by Skidmore Owings & Merrill (SOM), completed in 1960, was significant because of its embrace of this type of architectural modularity. The floor plan of the 52-story building, designed to house some 5,000 Union Carbide workers, featured a continuous run of carpeting and a ceiling comprised of a grid supporting lighting, air conditioning, and a Hauserman movable partition system.⁷ SOM's design allowed Union Carbide to rearrange the walls freely along the grid expanding and moving offices as needed within the modular system.⁸ Yet, Union Carbide found that despite the emphasis on flexibility that was built into the original design, making changes to the space was quite cumbersome and even expensive. According to Kevin Roche's project statement for the planning of their new suburban headquarters, Union Carbide was spending an estimated 1.5 million dollars a year in the mid-1970s (the equivalent of nearly 7 million dollars today) just on moves and reorganizations to accommodate regular status changes in their SOM-designed building in New York City.⁹ While the total number of moves represented by that figure are not specified, the Chemical and Plastics division reported that in 1976 alone 740 out of 888 workers in their area experienced a move or change to their space with around 20–30 percent of those changes requiring some form of new construction. Leadership from that division indicated that it cost around 150 dollars per workspace when no construction was required and an additional 10 dollars per square foot (30 cm) with construction costs.¹⁰ Those numbers combined with the total figure regarding moves in a year suggests that Union Carbide was moving thousands of workspaces a year in their New York City headquarters. Part of the expense associated with even small space changes was that nearly every change, for example, enlarging even a small 10 feet by 10 feet (304 cm by 304 cm) office to accommodate a promotion created a domino effect on the run of offices adjacent to that office in order to maintain space standards. Although the concept of modularity in postwar architecture provided some flexibility to organizations, it was also bounded and inhibited by the internal space standards that imposed limits on how the space could be changed. Thus, one of the central limitations of the Union Carbide building was not just the technical mechanics of the interior partitions, which were of course cumbersome and expensive to move, but also the set of rules or conditions under which those walls were required or allowed to move.

The modular interiors of postwar corporate architecture, however, were an important precursor and even a prerequisite of the open plan itself which depended on a conceptual shift in thinking about the relationship of the building to its interior arrangement. In their 1976 book, *Planning Office Space*, Francis Duffy, Colin Cave, and Frank Worthington differentiate the "shell" of the building (essentially the architectural envelope) from its interior scenery (furnishings). They argue that in the modern office, these two building elements, though interconnected, had very different life spans and purposes, with the shell serving

as a more permanent exterior of the building, and the scenery reflecting the shifting and evolving short term needs of the building's occupants.¹¹ In the open plan office concept, the shell of the building ideally offered an undifferentiated expanse of office space that could be freely arranged based on the needs of the organization and its workers.¹²

As the Quickborner Team's office landscaping and Herman Miller's Action Office were just gaining notice in the American architectural press in the late 1960s, E. F. Hauserman Company realized that their full-height movable partitions, though still in demand, were out of step with the emerging interest in these new office design concepts. The metal partitions that Hauserman had made for decades were movable, but they did not provide the kind of granular adjustability and instant flexibility associated with the office landscape and the emerging classes of systems furniture like Action Office. Recognizing the rapidly changing market, Hauserman began reimagining its product line to better address this new approach to office design and this new expectation of instant and constant change. In 1969, they produced a new brochure titled, "A New Concept to Manage Change in Business Interiors," which described how their low-height Divider Wall system in tandem with their full-height Ready Wall system could be used to create new landscape or open plan arrangements providing

the flexibility to change to the office layout you need to improve work flow and communications, to use space more efficiently, to experiment with new management techniques or departmental organization. And the change is not irreversible. If you find the previous layout was better you can go back to it ... easily and without incurring heavy labor costs.¹³

There was no specialized labor or additional costs required to make a change. Hauserman also developed a small line of accessories designed to turn their vertical partitions into active surfaces for projecting images, making notes with a water-soluble marker, supporting a chalkboard or corkboard, or even adding shelves. According to the Hauserman brochure, "Your thinking, your plans are not tied to the structure. You can change. You can manage change."¹⁴ Using Hauserman's new partition concept, companies were invited to try on a new arrangement, tinker with the organization, and explore new approaches to management in an inexpensive and quick manner without being locked into those changes. Hauserman thus characterized the office as a kind of sandbox of experimentation, and their partition system was a toolkit of flexible walls that would allow for effortless and ongoing organizational and spatial change.

Hauserman's fixation on the value of supporting organizational change was not just a marketing ploy. At the time, the Hauserman Company was in the process of implementing organizational change to help their own company adapt

to this rapidly shifting marketplace that no longer wanted full-height partitions and conventional office design. In their internal memos and meeting notes, Hauserman's leadership referred to Douglas McGregor's Theory Y concept and talked about the Drucker concept of "management by objectives" in which workers set their own individual goals, which were tied to larger organizational priorities. In one meeting from 1968, a management team at Hauserman described how they wanted to: (1) "be a growth company"; (2) "grow through managing change and finding the opportunities this presents"; (3) "manage change with people who are constantly growing in her capabilities"; and (4) provide "a climate of challenge for people to grow in."¹⁵ In transforming their product and promoting it as a tool for organizational change, Hauserman was reflecting their own organizational transformation that was underway at the time. As they describe in a 1970 annual report, "Some say there is nothing permanent except change. We at Hauserman agree wholeheartedly. The changing form of your business—your changing needs for space—represent our business opportunities."¹⁶ For Hauserman, organizational change was not just a pie-in-the-sky idea or a marketing gimmick, but something they were actively seeking for their own organization as they sought to take their company in a new direction, away from manufacturing building components (movable walls) and into the new systems furniture industry.

By 1971, Hauserman was already working on prototypes for their new office furniture concept when they invited a relatively new workplace design research group called Buffalo Organizations for Social and Technological Innovation (BOSTI) to consult on their new project. Following their initial consultation, Hauserman asked BOSTI to produce an analysis specifically looking at the problems of change in work environments, creating a set of requirements for a productive office, examining the economic issues for "high performance, adaptable office interiors," and identifying specific ways in which the new Hauserman product could respond to changing needs.¹⁷ The resulting report titled *The Management of Change for Productive Interiors* explored in detail the emerging problems of change in contemporary workplaces and the ways in which office space and office furniture could facilitate and support change in an active and ongoing manner. BOSTI concluded their report with eight practical design requirements that emerged from their findings. Drawing on these concrete recommendations from BOSTI, Hauserman's design team created their new Office Interior System (OIS) furniture to facilitate and support ongoing organizational and spatial change.¹⁸

From the outset, Hauserman conceptualized their systems furniture as an adaptable material intermediary between their human and organizational inhabitants and the architectural spaces they inhabited. The promise of instant and constant flexibility for change was embedded in the design, and was central to the ways in which the new furniture was marketed and sold. Hauserman was

just one example, but their investment in this idea of change as an organizational value and their emphasis on change in the development of the OIS system illustrates how these ideals of organizational change and the possibility of change, as conceptualized in this new class of office furniture, were interconnected. Of course, Hauserman was not unique in this regard. Many other furniture companies in the late 1960s and early 1970s were similarly trying to upend their own product lines to compete in this emerging market of systems furniture, and like Hauserman, companies like Westinghouse and Steelcase highlighted the easy flexibility and adjustability of their furniture, which was designed to support and facilitate ongoing space change.

Systems Furniture and the Materiality of Change

Early manufacturers characterized this new class of open plan furniture as a modular system of components that would provide organizations infinite flexibility to make changes to their space and to reflect organizational changes. In practical terms, this new class of office furniture was a collection of individual components including writing surfaces, storage, panels, and shelves that organizations, architects, designers, and users could assemble and combine in a variety of different configurations. The modularity of systems furniture and its emphasis on supporting change was dependent on visual continuity among the various parts and pieces of the system.¹⁹ The desks, storage, and panels all had to coordinate visually (as well as technically) to ensure that the image of the office remained attractive and professional regardless of the configuration in use. In fact, this concept of visual continuity in office planning was itself an expression of modularity in that it reflected the coherent application of materials, finishes, and components that could easily mask the many iterations of change. Aesthetic coherence ensured that the various parts of the system would always fit together and would be visually attractive and complete no matter how they were assembled (Figure 9).

In marketing materials, systems furniture often included assembly images with a hand using a standard tool to assemble a panel; for example, Figure 10 shows a hand tightening a screw on a Steelcase Movable Walls panel. This kind of image reinforced the ideal of easy assembly. Yet in practice, the technical aspects of the system were actually quite complicated; Figure 11 shows a detailed technical drawing of a Steelcase Movable Wall panel and all of its parts and pieces. In general, panels by most systems furniture manufacturers required “connectors,” a type of post that linked the panels to one another. Top caps and end caps gave the partitions a finished appearance (disguising the connections and the unfinished ends of the panels), and “glides” added to the feet supported



Figure 9 A technical drawing of Herman Miller's Action Office 2 from 1972 illustrating the modular design of the components. As can be seen in this image, all of the components of Action Office were stylistically similar and attached by simple connectors and basic hardware to facilitate easy assembly and change.

Source: Courtesy of Herman Miller Archives.

the panel, allowing it to move and adjust to accommodate uneven floors. Writing surfaces attached to partitions by brackets, cantilevers, or stretchers, and the drawers and other storage elements mounted underneath the writing surfaces or connected to the panels directly. At an even more granular level, each of these parts were attached to each other using very small standard hardware such as screws, nuts, and bolts, each of a particular size and weight necessary for each component. A survey of thirty-seven different lines of furniture from 1974 illustrates the sheer variety of connectors across the various lines describing complex systems of brackets, bolts, screws, clips, hooks, key holes, and even specialized (manufacturer specific) hardware.²⁰ If this list of hardware used to assemble systems furniture is dizzying to read, just imagine the challenge of actually assembling this furniture with all of these parts and all of their fussy hardware.

Because each system was designed to allow multiple configurations, some components might have several different ways in which they could be connected, with each iteration requiring its own set of specialized hardware. In every systems furniture line, the different component pieces offered a number of possible configurations, and as a result, ordering systems furniture

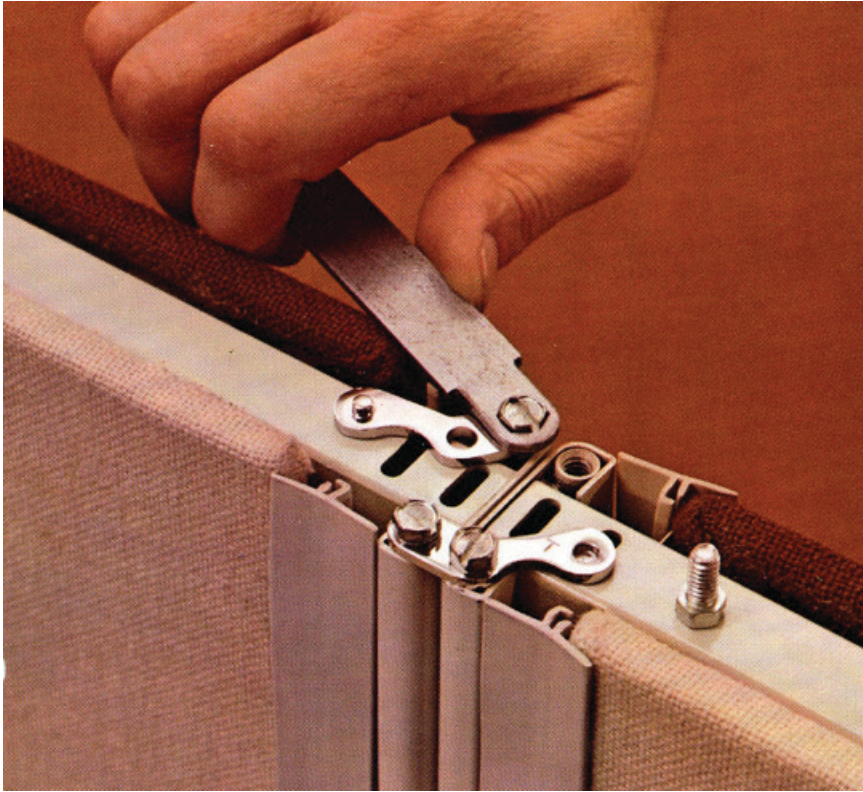


Figure 10 Image of a hand screwing the top of a panel in Steelcase's Movable Wall system from 1976. This kind of photograph with a close up of a hand attaching a panel was a common type of image in systems furniture brochures in the 1970s.

Source: Steelcase Corporate Archives.

was sometimes incredibly complicated requiring careful planning to ensure that the correct hardware was on hand to support the planned installation. Knoll's Stephens line had different hardware "kits" depending on the particular connecting workstations and arrangements of panels and work surfaces, and those kits had to be specified as part of the original furniture order.²¹ If one type of kit was used in the original installation, and a later space change required a different hardware combination, that new kit would need to be ordered for the planned change to be implemented. Systems furniture was often portrayed in the furniture catalogs as an adaptable, flexible, and infinitely changeable system, but the materiality of that change was constrained by particular configurations and options. In other words, manufacturers designed the components to connect in predetermined ways; the furniture thus prescribed certain types of installations and inhibited (or even precluded) others.

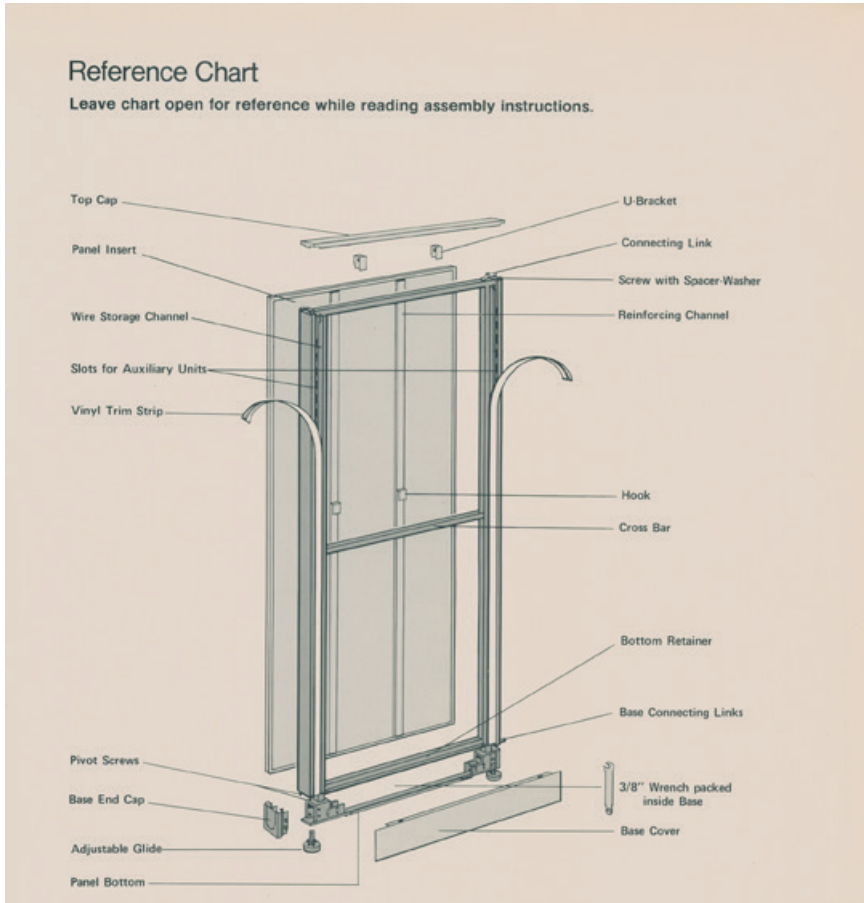


Figure 11 Detailed technical drawing of a panel assembly for the Steelcase Movable Wall system from the 1970s. This illustrates the basic components and hardware of a standard Movable Wall partition; note that Steelcase packed a special wrench for assembly into the base of the partition for shipping.

Source: Steelcase, *Assembly Manual Steelcase Movable Wall System* (n.d.). Trade catalog from the William Pahlmann Papers Series 25 Box 29 at the Hagley Museum and Library and used with permission of Steelcase Corporate Archives.

Systems furniture components, though designed to integrate with other pieces in the same system, were not generally well suited to mix with other unrelated systems. The parts of Knoll's Stephens System could not be easily intermingled or connected with Herman Miller's Action Office, Hauserman's OIS, or Steelcase's 9000 line. Not only did the various systems have their own finishes and fabrics, making them visually and materially distinct from each other, they were also technically distinct; a work surface from one system could not be easily connected to a panel from a different manufacturer.²² By choosing a

systems furniture line, organizations were making a long-term investment with a particular manufacturer, and even after their initial purchase of furniture for a new installation, organizations would often have to purchase additional components and parts to augment their inventory over time. As a result, long-term support and ongoing production and expansion of the system (with new parts to accommodate new needs) were critical criteria for organizations in choosing a systems line.²³ The open plan office was often described by manufacturers as a system designed for change and flexibility, but organizations adopting these open plan systems were immediately locked into a certain furniture line with specific components and predetermined configurations.

Just as the systems lines could become a limitation over time for customers who had adopted a certain type of furniture, the system also imposed constraints on the manufacturer. As systems furniture manufacturers expanded their furniture lines to offer new types of installations and new amenities, they needed to integrate those new parts with the system and ensure that organizations could use the new components with their older generations of furniture. In other words, manufacturers were also locked into their systems as they needed to continue to support their previous clients (who still needed to purchase furniture components for their offices) while also staying competitive in the market as they acquired new clients and addressed new needs and emerging problems. For example, in the early 1970s, integrated lighting systems were a custom concept in office furniture, but by the mid-1970s many of the major furniture lines had added their own integrated lighting systems. These newly designed furniture lighting components often featured a combination of ambient “up lighting” (with lighting elements that created reflected light on the ceiling) and downward task lighting at the workstation itself that could be added post-installation if necessary.²⁴ When Steelcase created a new lighting component for their systems furniture in 1977, they ensured that it would work with either of their systems lines to allow their legacy clients to retrofit the new lighting system to their older furniture.²⁵

Perhaps nothing was more challenging in the logistics of the open plan and systems furniture design than the provision of power. Early systems furniture relied on architecture for adequate access to power and phone lines, but over time this dependence on the architecture for essential utilities became a barrier to change. As architectural journalist Donald Canty described, this issue of providing access to power and telephone cables served to effectively “anchor work spaces” ultimately inhibiting much of the purported flexibility in systems furniture.²⁶ Part of the reason this became so problematic is that even through the 1970s, most American office buildings were still being designed and constructed with relatively modest power and communication access intended for general office work (typewriters, telephones, and other such common devices). As I will discuss in more detail in Chapter 5, the proliferation of “power-hungry” equipment, particularly computers, required architects, designers, and organizations to give

more consideration to the problem of electrical distribution in the office.²⁷ To address this growing problem of power and communication access in the office, systems furniture began to evolve over the course of the 1970s from light-weight partitions into an architectural power grid providing universal office access to electricity, communication cables, as well as lighting.

The earliest iterations used auxiliary raceways, channels mounted on the front or bottom of a partition accommodating runs of cables for power and communication. For example, both the Steelcase Mobiles and Steelcase 9000 lines in the 1970s had a means of integrating cable runs within the furniture in order to make power more accessible, but also to keep the visual clutter of power and telephone cords out of sight.²⁸ This type of furniture was not actually “powered,” it was merely a conveyance of power, in some instances even relying on a daisy chain of extension cords to connect to a floor, wall, or ceiling power outlet. It was not uncommon for multiple workstations to rely on power access from a single outlet or tombstone in the 1970s; designer John Pile describes “stringing wiring” through raceways and plugging in clusters of workstations into a shared outlet.²⁹ This method of handling wiring, using runs of extension cords, was not only poor in terms of power issues (creating overloaded circuits) it was also a fire hazard that was even illegal according to some local electrical codes.³⁰ Any change in office design often required an electrician to manage parallel changes to the power thus limiting the purported flexibility of the furniture significantly.³¹

In 1976, furniture company Haworth released their ERA-1 system, which created the first powered (electrically pre-wired) partition system. In an advertisement for the new powered system, Haworth invited architects and designers to “plan your spaces for the people, not the building.” Featuring internal wiring, compartmentalized raceways, and flexible connectors that “snapped” into place completing a circuit, the ERA-1 represented a fundamental shift in thinking about the provision of power in the office.³² Soon the other major furniture manufacturers were scrambling to create their own integrated powered components to compete with Haworth’s revolutionary system.

Haworth’s ERA-1 transformed systems furniture into a critical part of the power infrastructure of the office, and by the early 1980s all of the major furniture manufacturers boasted their own integrated power elements designed to support both communication and power needs. Yet, specifying and installing these powered elements only made the systems furniture planning more complicated. A 1988 planning guide for the CETRA system by furniture manufacturer Kimball offers exhaustive detail of the important considerations and caveats in planning powered furniture from identifying the source of power (ceiling, floor, or wall) to determining the placement of the raceway in each panel. To even connect the electrical supply from one panel to the next, designers had to specify the jumper cables that would be used, and those could vary based on both the angle of the

connector or (straight, 90°, 120°, 135°) as well as the size of the panels.³³ The integration of powered elements in the design of systems furniture thus made the furniture itself much more complex in its design, much more challenging to specify in a given project, more complicated to install, and increasingly difficult to change.

In fact, by the 1980s, as systems furniture lines were becoming more architectural in their functionality and form, some new furniture lines embraced this idea of the furniture as interior architecture. In the release of their new furniture line Ethospace in 1984, Herman Miller used the language of architecture to describe the new furniture: the partitions were “walls,” the glazed panels were “windows,” and the interior arrangement was called a “skyline.”³⁴ Similarly, Haworth augmented their Places line with a new line of “Architectural Elements” including fanlights, French doors, gabled panels, and various muntin options (window panes) creating a distinctly architectural aesthetic for the workspace (Figure 12).³⁵ Buildings in turn became even more skeletal providing only an empty shell of an office with basic power and heating ventilation and air conditioning (HVAC), while increasingly relying on systems furniture to create the entire interior



Figure 12 Image of a Haworth Places workstation which shows the integration of architectural elements into systems furniture. Note the panel featuring window panes with rectangular muntins and a pediment-style top.

Source: Image from Haworth, *Places*, 1989. Trade catalog from the Hagley Museum and Library and used with permission of Haworth.

arrangement, and provide lighting, power, and communication cabling for all workstations.

Facilities Management and the Logistics of Organizational Change

The consistent marketing of this new open plan furniture as a flexible “system” designed to support and facilitate change suggests that organizations adopting the open plan valued that attribute. Obviously, the architecture and design press and office furniture manufacturers regularly discussed this idea of flexibility in the new open plan and systems furniture, but the business and mainstream press described the benefits of the open plan and systems furniture in very similar ways. An article from the business journal *Administrative Management* from 1973 describes how the new “modular” workstations were only of value if organizations installed them in a manner that allowed some flexibility in their arrangement enabling change and allowing for growth. Describing the practical uses of this new type of systems furniture, the article described how organizations could arrange the furniture in a variety of configurations, and emphasized that the components were interchangeable, allowing organizations to easily swap out elements of the workstation and quickly add on additional workstations as needed.³⁶

The promise of instant and constant space change to support organizational change was appealing at the time in part because public and private organizations alike were navigating an enormous amount of change in real time. The new theories of management at the time promised to create more engaged and productive workers but required a parallel cultural change. New organizational structures promised to make the organization flatter and more dynamic, but that often required a parallel reorganization of the company and its divisions. New office technologies, particularly computers and automation, promised increased efficiency and productivity but required entirely new ways of organizing and structuring work. Meanwhile, the economic climate in the United States was also rapidly changing from the seemingly endless growth and expansion that characterized the 1960s to the belt tightening recessions of the 1970s and 1980s that required still further organizational change to stay competitive.

Private organizations were typically tight-lipped about the nature of organizational change underway, but public organizations were often quite transparent, documenting, referencing, and discussing organizational change in countless memos, letters, and reports. In the State of Wisconsin there were numerous organizational changes through the 1970s and 1980s; the Department of Administration alone had five major reorganizations between 1968 and 1979. From 1978 to 1989 there were more than twenty different formal reorganizations across various agencies throughout Wisconsin State Government, many in

connection to proposed space changes.³⁷ In addition to formal reorganizations and programmatic changes, other types of change were also underway at the State including efforts to decentralize certain activities (personnel and payroll), and efforts to centralize others (word processing and data processing).³⁸ Many of these kinds of organizational changes truly did require a space change, from just moving around a few workers to creating or dismantling entire organizational divisions. In this context, it is not surprising that the open plan's promise of easy and inexpensive space change resonated with business executives and management at the time.³⁹ For organizations, the office itself was a tool for change, facilitating the shifting around of people and divisions, supporting the creation of new teams or departments, and reflecting the cuts and expansions of staff. Public and private organizations alike were strongly attracted to this purported ability of the open plan to make change quickly, cheaply, and with minimum disruption in part because they believed that the design of the office should reflect the design of the organization.

Early reports about the open plan often cited the huge percentage of moves and changes or space "churn" for the companies in the months and years following its adoption. In their 1977 story about the Weyerhaeuser open plan offices, *AIA Journal* indicated that roughly a third of Weyerhaeuser's 900 workers were moved each year, and some workers even said that they had been moved three to six times since their move into the new building in 1971.⁴⁰ A report from *Progressive Architecture* from 1977 found that American advertising agency Young and Rubicam made changes to their open plan offices every sixty to ninety days, and IBM estimated that some 25 percent of their work stations were relocated each year in their corporate headquarters.⁴¹ Similarly, an *AIA Journal* story from 1980 reported that Citibank in New York City moved 20 percent of their workers each year, and in some divisions the move rate was as high as 50–75 percent.⁴²

Organizations also used the open plan as a tool for implementing space reductions. For example, in the early 1980s the Department of Administration at the State of Wisconsin had to reduce their office space by an additional 12,466 feet (approximately 3,800 meters) just a few years after moving into their newly designed open plan office.⁴³ The description of seemingly unending space changes in the State of Wisconsin from the late 1970s through the early 1980s reads like an elaborate game of Tetris in which the various units, departments, and agencies were "shoehorn[ed]" into spaces with the express goal of squeezing as many people as possible into the available space.⁴⁴ This kind of squeeze on office space was not unique to the State of Wisconsin; in the words of one space planner in 1979, "all employers want to know is how many people they can jam in there."⁴⁵

These ongoing changes were rarely handled by the architect or designer who created the original layout.⁴⁶ Instead, private and public organizations

increasingly relied on an internal facilities management unit to handle these regular space changes. While the idea of having a dedicated space manager or administrator was not really new, the field of facilities management (and their associated responsibilities) grew significantly in the 1970s and 1980s, at least in part because of the widespread adoption of the open plan and systems furniture that required a new approach to managing and coordinating office space.⁴⁷ As noted above, systems furniture was marketed by manufacturers as “easy” to change and move, but in reality it was a complex and messy collection of components and hardware that had to be managed and maintained. Even a single workstation might have fifty or more components associated with it, and a large corporate office might have hundreds or even thousands of individual workstations that needed to be managed and maintained.

In 1978, a facilities manager at Hallmark told Herman Miller that managing the growing collection of parts and pieces of the furniture system was one of the biggest challenges in using the Action Office system. They further describe how designing a new space with all new equipment was relatively easy to plan and implement, but the changes over time were much more complicated due to the continuous shuffling and repurposing of the system components. This dynamic aspect of the system made the inventory process not only more challenging but also even more essential. Frequently the components necessary for a planned change were not on hand, creating delays as they scrambled to locate or order the necessary pieces.⁴⁸ Every space change therefore required a careful accounting of the furniture on hand in order to make a new layout, and creating an in-house inventory system to track those pieces became essential. The furniture inventory was a functional necessity in the planning process, and facilities managers were often tasked with creating their own inventory system from scratch, which required the creation of an internal tagging system for the parts that would track the identifying information of each component (including its manufacturer, its systems furniture line, its part number, and even color) as well as its physical location whether in storage or on the floor in a specific workstation. Once they had created a working inventory, facilities managers had to constantly update their records to keep track of the growing collection of disparate furniture parts and pieces as they moved around the office.⁴⁹ Each furniture component in the office had a double life—simultaneously existing in physical form in a workstation or in storage, and as a piece of data in an inventory. The inventory of systems furniture was a critical backbone for planning and implementing space changes.

Space changes were also challenging for facilities management in part because requests for change often arrived with little notice or forewarning. Units requesting a space change frequently expected a very quick turnaround, and typically required a change to a space that was already occupied by a group of workers, making the planning more complex.⁵⁰ In other words, the planning not only involved rearranging the space and the furniture to support the necessary

change, it regularly included shifting a whole group of workers already in place to accommodate whatever change was necessary. The architectural space could also impose limits and constraints as older buildings and spaces had to be adapted over time to address new needs, requirements, and expectations.⁵¹ Oddly shaped spaces, inconveniently placed columns or service elements, limited power or lighting access, and other architectural issues could inhibit the planned arrangement in unexpected ways and created regular logistical and technical challenges for facilities managers.

Organizations often had multiple lines and generations of furniture in use and were sometimes using those furniture components in ways that belied the notion of a complete integrated system. This was not an unusual issue—at the time, organizations often adopted systems furniture in a gradual way, starting with a small test case and slowly updating new divisions and areas with the new type of furniture. As a result, new systems furniture would frequently sit side-by-side with older furniture types and even different makes and models of furniture.⁵² In an era of tight budgets, facilities managers often tried to reuse older furniture as much as possible, and so the patchwork quality of systems furniture would only become more pronounced over time as space changes required reconfigurations, and facilities management teams had to creatively repurpose parts and components that were not necessarily designed to fit together.

At Purdue University, the office plans for Freehafer Hall from its first construction as an “office landscape” in 1971 through the 2000s demonstrate the complexity of office redesigns even in a purpose-built open plan. When it was first constructed, Freehafer Hall used simple office landscape furniture with freestanding curving screens, freestanding storage, and freestanding desks. There were many iterations of plans from 1971 to 1986 suggesting that making changes using the original “landscape” furniture (freestanding desks and screens) was fairly common and relatively easy. However, in 1987 and 1988, Freehafer underwent a major redesign (including a building addition) in which the university began integrating more furniture from various lines of systems furniture, some components that were purchased for Freehafer and other pieces that were acquired from other campus buildings. In contrast to the relatively simple office landscape layouts of the past, the plans from 1987 to 1988 include increasingly complex inventories of furniture from eleven different manufacturers. Purdue space planners had to juggle multiple systems furniture lines, specifying the various work surfaces, storage elements, and partitions as well as all of the hardware (screws, attachment kits, and brackets) necessary to hook all of the parts together.⁵³

This gap between the vision of instant and constant space change as it was portrayed by furniture manufacturers and the reality of implementation at the organizational level illustrates the problem of talking about change only in the abstract, as it was described in systems furniture brochures. Office

furniture materials of the period often depicted office design as a hermetically sealed system of perfectly integrated components that easily fit together in any configuration, but real offices often had multiple lines and generations of furniture that they were managing many of which did not go together at all. Changing the design of the office sometimes meant that facilities management teams had to find ways to connect parts and components of furniture that were not designed to fit together. The planning of office change could become increasingly complicated over time as organizations were mixing different lines of systems furniture, different generations of components, and adapting to changing work processes and technologies. Every change, even relatively modest ones, required careful planning, and the implementation of space change was often messy as organizations were negotiating changing organizational structures and changing technologies.

Workers' Perspectives on Change

In 1982, a press statement from a Wisconsin employee union described how a series of recent reorganizations, budget cuts, and layoffs had resulted in continuous shifting of office space. According to the press statement, in the previous year, 125 workers and 5 supervisors in one of the agencies were let go in tandem with a reorganization, which also resulted in an office change, followed by still further budget cuts, additional layoffs, and another reorganization requiring yet another round of office changes; the agency's "beleaguered workers began referring to this process as 'rearranging the deck chairs on the Titanic'."⁵⁴ Similarly, a worker at Blue Cross Blue Shield in New York City in 1977 described how their open plan office was like "an Erector set" in that "we could leave one night and come back to find all the cubicles gone. This is *the* modern office space, floating and subject to change. We have to forget that we, too, may be as movable as the office."⁵⁵ At the organizational level, change was often conceptualized through new policies, processes, technologies, office spaces, systems, budget cuts, layoffs, and reorganizations, but these diverse changes were not mere abstractions discussed in boardrooms and written on pieces of paper, they were real changes that affected workers' lives in significant ways.

Looking at change through the lens of workers thus offers an altogether different perspective on the nature, meaning, and experience of organizational change in this era. The open plan enabled organizations like the State of Wisconsin and Blue Cross Blue Shield to enact organizational change relatively quickly, but workers had an altogether different experience of change. For those on the ground of these organizations, space changes were a manifestation of a state of continuous upheaval and economic uncertainty in which one's job could change or be eliminated altogether with the shifting of an office partition. For

workers, change was underway at every level from large-scale reorganizations to tiny unit-level changes that might only change a single job or a single position, and in some cases these small and large changes were all happening at the same time. In the words of one Wisconsin State worker from 1975: "In the year that I've been in state government: 1) the state planning program has changed; 2) the state planning staff has changed; 3) management in other agencies (with whom we have been developing 'credentials' and positive working relationship) has changed; 4) DOA leadership has changed; 5) DOA is changing; 6) I realize that I've stopped singing in the shower."⁵⁶ This ripple of change at all levels of the organization meant that workers were constantly navigating new organizational relationships and new responsibilities, alongside new systems, technologies, and processes.

In their 1978 Steelcase study, Louis Harris and Associates found that the majority of office workers surveyed (including clerical, secretarial, professional, and executive workers) had seen some change including changes to office space (location of workspace or office redesign), organizational changes, and change in the structure of one's job requiring new skills, new tasks, or new responsibilities.⁵⁷ These kinds of changes were not neutral from a worker's perspective; they could potentially improve the quality of working life, but they could also make work more stressful. In an overworked, overstretched unit, adding some new workers to help relieve the work flow would certainly be a benefit, but a cut to an already overworked unit could make the work process even more stressful as the remaining workers took on more responsibility to cover the lost labor. In other words, talking in broad general terms about organizational change misses the textures and nuances of change, and the ways in which workers in different divisions and positions within an organization experienced that change.

Perhaps the most significant change in this period was the huge growth in unemployment through the 1970s and early 1980s. From 1973 to 1975, the number of unemployed workers in the United States doubled, and among the unemployed in 1975, 58 percent were individuals who had lost their jobs.⁵⁸ It was common at the time for organizations to treat clerical staff as expendable, and as a result workers in those positions were often the most vulnerable to cuts. A 1973 report titled *Work in America* found that low-level, white-collar workers were viewed as nonessential, and were therefore often the first jobs to be cut in an economic downturn.⁵⁹ Women and workers of color, even those in professional roles, were frequently among the first to lose their jobs as a result of their low seniority.⁶⁰ Meanwhile, as they made drastic cuts to their full-time staff, many organizations increasingly relied on a growing class of contingent labor in the form of part-time and temporary workers.⁶¹

Layoffs often came in tandem with other changes to the organization, which created additional pressures, challenges, and problems for workers. In their

1984 survey on worker stress, the working women's union 9 to 5 found that nearly 60 percent of respondents had seen some staffing reduction in their offices, 50 percent had seen a "freeze on salaries," 67 percent had experienced some kind of "speed up" in their work, and 60 percent indicated that they had either been promoted or given more responsibility.⁶² Among workers who had experienced staff cuts, pay freezes, and speed-ups, there were higher rates of reported health issues including increased headaches, nausea, insomnia, physical pain, and depression.⁶³ Thus, staffing cuts and layoffs came side-by-side with other efforts to increase efficiency and reduce costs, and the attendant pressure could even manifest through feelings of physical stress.

New technology was another ongoing source of change and disruption in the structure, culture, and process of office work in the 1970s and 1980s. Organizations often characterized "office automation" as a means of increasing operational efficiency, but it often included dramatic changes to the structure of work for clerical staff and frequently brought significant staffing cuts and layoffs in its wake. In 1975 when Citibank implemented a new automated system for their letter of credit division they cut their staff from 142 workers to only 100.⁶⁴ Many office automation efforts from the 1960s through the 1980s in both public and private organizations also required a shift of clerical work from individual units, offices, and divisions, into centralized clerical units such as word processing or data processing. It is worth noting that centralizing administrative tasks was not a new idea. Throughout the twentieth century, pools of workers had often been used to staff certain types of clerical divisions such as stenography, typing, and keypunch, but the emphasis on adopting automation in tandem with centralization through the latter part of the twentieth century significantly changed the structure of clerical work.⁶⁵

By plucking the mostly female clerical workers out of their home divisions and placing them in newly re-centralized clerical units, organizations of the 1970s and 1980s treated clerical workers as not only "movable" in a physical sense, but essentially interchangeable within the organizational machinery. Organizations treated low-level workers as a portable unit, whose body and labor could be easily slotted in anywhere it was needed, just like the systems furniture workstation they inhabited. Underneath this way of thinking was the assumption that clerical workers were all doing the same types of work, had the same essential responsibilities, as well as the same knowledge and skills. Yet, as Shoshana Zuboff argues, clerical workers often had specialized knowledge of the work in their particular division, and played important roles in the overall functioning of the division.⁶⁶ The adoption of automation often replaced these specialized knowledges and skills with computerized processes that stripped away the variety and complexity of clerical work, including its social aspects. Tasks that formerly required a worker to actually talk to another worker were now often handled through a machine interaction or workflow instead of through a

human interaction.⁶⁷ In word processing units, workers were socially and spatially isolated, working in windowless rooms far away from the regular staff, and often unknown by the professional workers whose letters and reports they typed. As Anne Machung describes, this approach to word processing “destroys much of the social fabric that formerly wove the office together, undermining bases of loyalty and interaction between clerical and managerial staff, dividing their social worlds from each other, and further segmenting these relationships, all in the name of productivity.”⁶⁸

Ironically, even as many companies in this period embraced the ideals of McGregor’s Theory Y management in their upper ranks, they implemented automation in their clerical units in ways that reinforced a Theory X approach, controlling and restricting workers instead of giving them autonomy. Technology engineers designed these new automated systems such as word processing and data processing to sharply limit workers’ ability to operate within the system, and even monitor workers by tracking and recording strokes. These automated machines constantly tracked the work of their operators, tallying every line, page, and document, and reporting those statistics as direct measurements of productivity. In some cases, these new systems were even set up to “beep” at workers if they were not meeting their productivity quota.⁶⁹

Not surprisingly, clerical workers were sometimes resistant to these organizational and technical changes in the structure of their work. In 1979, as the State of Wisconsin was encouraging more agencies to adopt a centralized structure for clerical tasks like word processing, some workers expressed real concern about this proposed change. One division head in the State of Wisconsin described the reluctance of their clerical staff to be assigned to a centralized word processing unit, writing in a memo: “to a person they have no desire to be in an ‘assembly line’ production role.”⁷⁰ Workers frequently viewed these automated word processing units as a factory-like spaces akin to working an assembly line. The work was also stressful, tedious, and alienating, and as a result these newly automated jobs often had very high turnover.⁷¹ In some word processing centers, the turnover was reportedly as high as 100 percent a year as operators sought better working conditions and better pay.⁷² Workers in the State of Wisconsin Department of Administration word processing unit in 1985 had such low morale, that the state brought in a faculty member from a local college to work with the staff to help improve the situation. In the evaluations from those sessions, many of the participants described the negative aspects of their work including their feelings of unhappiness in their jobs, high levels of stress and anxiety, a “dehumanizing” work culture, frustration with the way in which they were treated by workers outside of their division, and a general lack of collegiality among workers in the unit.⁷³ These types of negative experiences in centralized word processing and data processing units were well-documented in both public and private organizations through the 1970s and 1980s.⁷⁴

The physical environment of the office in the 1970s and 1980s was also becoming less comfortable for workers as a result of changing building systems. Concerns about energy consumption and waste beginning in the 1970s meant that many of the new buildings constructed in this period featured building systems designed to minimize energy usage. As Michelle Murphy describes, new ventilation standards released in 1974 sharply reduced the intake of fresh air creating stifling and unhealthy office spaces. The lack of sufficient fresh air in office buildings was only made more problematic by the wide-spread practice of allowing smoking in office spaces in this time period.⁷⁵ Toxic and hazardous chemicals and materials in the construction and furnishings of office buildings along with pollutants generated by new machines like copiers created still further potential health hazards for workers.⁷⁶ Even temperature levels were increasingly controlled based on energy efficiency standards rather than on worker comfort. In the early 1970s the US federal government established standard temperatures for all government office buildings in order to ensure energy efficiency. Describing this shift in policy in 1973, the General Services Administration (GSA) described how “halls have become a little darker, and working spaces became suddenly warmer in the summer and cooler in the winter” to address concerns about energy consumption.⁷⁷ These concerns that were prominent in the 1970s foregrounded the engineering and construction of buildings, and gave little consideration to the comfort and well-being of the human bodies contained within.⁷⁸ Through the 1970s, the open plan was often characterized as a tool of increased energy efficiency because it eliminated the need to heat, cool, and light small enclosed spaces, and allowed for a more efficient air handling system.

These details about the ambient air quality in office buildings reveal the inner textures of everyday life in an office building in this period. While the ambient air quality might seem to be a micro-climate (a particular office building in a particular place at a particular time), the basic preconditions of that micro-climate were actually common in office buildings across the United States. This issue of air quality and ambient temperature, which were tied to concerns about cost saving and energy efficiency, were connected to the other changes in the economy, the organization, and technology. The early proponents of the open plan imagined a future of change that was dynamic and progressive ultimately creating a better organization, happier more productive workers, and a more supportive and responsive office, but the changes of the 1970s and early 1980s often produced negative outcomes for workers. The interweaving of organizational change and space change in the open plan meant that space was not just a reflection of organizational change, it was a spatial and material manifestation of change. Workers did not just experience the change in a general sense; they physically inhabited this climate of uncertainty and discomfort.

The Problem of Change

The descriptions of the open plan from its early promotion in the United States read like a love letter to the possibility and potential of radical organizational, social, technological, and spatial change. Architects, designers, and manufacturers saw the open plan as a flexible and nimble solution to the problem of constant change.⁷⁹ In the late 1960s, advocates of the open plan saw organizational change not only as inevitable, but also as an intrinsic part of the natural cycle of growth and creative innovation. Underneath this idealistic vision of the open plan was a sincere belief among furniture manufacturers as well as architects and designers that every change in an organization should have a parallel change in office space.

This fantasy of change was central to the design and marketing of systems furniture, which promised that the new furniture could readily adapt to the ever-shifting needs and challenges of organizations and their workers. Despite the marketing claims, in practice, systems furniture was not so much a perfectly integrated and interlocking system of parts, but rather a messy collection of disparate components and hardware that required careful planning to design and assemble and even more careful planning to manage and change. Furthermore, systems furniture became more complicated over time as manufacturers updated and expanded their systems to support new needs. Advocates of the open plan envisioned change as an intrinsic part of the concept and systems furniture was designed as a modular toolkit that would be continuously updated, changed, and augmented to address emerging needs. Yet those changes also made the system itself increasingly complex to manage and use, particularly for facilities managers and their teams who were responsible for managing the furniture and the never-ending space changes.

For workers, this landscape of continuous organizational and space changes in the 1970s and 1980s became an outward manifestation of the instability of their own jobs, which could be changed or eliminated at any moment. Even workers in high level positions within organizations were increasingly vulnerable to cuts and layoffs, but this increasing pressure on workers was not carried by all workers equally. Clerical staff, women, and workers of color were often the most vulnerable to cuts, layoffs, reorganizations, and technology changes that could transform or even eliminate their jobs with little warning or notice. While those promoting the open plan in the 1960s and 1970s often characterized change as an abstract concept, it had real world implications and consequences for workers whose positions, bodies, and needs were often seen as expendable to the needs and goals of the organization.

